

BITSAT College & Branch Predictor: Pitch & Flow Documentation

This document pitches the design strategy, value proposition, and under-the-hood engine architecture of the **BITSAT College & Branch Predictor**.

The Pitch: Why This Predictor Wows Users

Traditional college predictors are clunky, rely on static tables, and fail if their backend databases are temporarily unavailable.

Our **BITSAT College Predictor** is engineered as a modern, premium **MERN Stack Dashboard** that solves these limitations:

- Dynamic Glassmorphic Interface:** Dark aesthetics featuring smooth sliders, radial index gauges, and neon color-coded chance badges that update instantly as the user types.
- Reliable Dual-Engine Flow (Offline Resiliency):** If the backend MongoDB is unreachable, the frontend seamlessly runs the *exact same* mathematical prediction algorithm client-side, showing a **Local Engine (Offline Mode)** badge instead of breaking.
- Multi-Year Trend Analysis:** Displays cutoffs side-by-side (2022, 2023, 2024) to give applicants context on how competition has grown.
- Probability-Based Status Badges:** Uses an adaptive cutoff matching algorithm to show real chances rather than a simplistic "Yes/No":
 - Very High (95%+ chance):** Score is $\geq \text{Cutoff} + 15$
 - High (75% - 95% chance):** Score is between Cutoff and $\text{Cutoff} + 15$
 - Medium (40% - 75% chance):** Score is between $\text{Cutoff} - 10$ and Cutoff
 - Low (10% - 40% chance):** Score is between $\text{Cutoff} - 25$ and $\text{Cutoff} - 10$
 - Unavailable:** Score is below $\text{Cutoff} - 25$

Architectural Flow & Logic

The system flows through a deterministic cycle from input to rendering:

```
graph TD
    A[User inputs Score or adjusts Filters] --> B[Fetch http://localhost:5000/api/predict?]
    B -->|Success| C[Express Backend Processes Request]
    B -->|Connection Fails| D[React Local Engine Runs Client-Side Matcher]

    C --> E{MongoDB Active?}
    E -->|Yes| F[Read Cutoffs from Cutoff Model]
    E -->|No| G[Fall back to server-side local JSON]

    F --> H[Apply Probability Classifier]
    G --> H
    D --> H

    H --> I[Sort Branches by probability desc]
```

```
I --> J[Render Campus-wise Grouped Cards]
J --> K[Update Radial 'Admission Chance Index' SVG]
```

The Classifier Formula (JavaScript Implementation)

To estimate the probability dynamically, the matcher compares the input score to the latest (2024) cutoff :

This ensures that even if a student is slightly below the cutoff, they see a realistic "Medium" or "Low" chance rather than an outright rejection, mirroring real-world counseling iterations.